

Matas Vitkauskas

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Education

McGill University, Montreal, Canada

Aug 2025 – Present

PhD in Human Genetics

Thesis topic: Single-cell spatial genomics of early life adversity and depression.

Yale-NUS College, Singapore

Aug 2018 – May 2022

Major: Life Sciences; Minor: Physical Sciences (Chemistry)

CAP: 4.47/5.00; Major CAP: 4.79/5.00

Publications

- Xie, Jessica Jiabin*, **Matas Vitkauskas***, Quyen Do, et al. 2026. “Single-Cell Transcriptomics Revealed Molecular Vulnerability in a Human Midbrain-like Organoid Model of Parkinson’s Disease.” *iScience* 0 (0). <https://doi.org/10.1016/j.isci.2026.114674>.
- **Vitkauskas, Matas**, and Ajay S. Mathuru. 2020. “Total Recall: Lateral Habenula and Psychedelics in the Study of Depression and Comorbid Brain Disorders.” *International Journal of Molecular Sciences* 21(18): 6525. <https://doi.org/10.3390/ijms21186525>

Awards and Honors

- Onni Foundation Award (2000 CAD).
- Lithuania Foundation Award 2026 (3000 USD).
- McGill Graduate Excellence Award (15,580 CAD).
- Yale-NUS Summer Research Program scholar 2019 (grant award 3,500 SGD).
- NUS Outstanding Undergraduate Researcher Prize 2020 (cash award 2,000 SGD).
- Harvard Summer Honors Undergraduate Research Program (SHURP) scholar (grant award 3,000 USD).
- 2nd place in 10x Genomics Spatial Transcriptomics Hackathon 2024 (prize award 200 SGD).

Research Experience

McGill University, PhD Student, Canada

Jul 2025 – Present

- PhD program in Human Genetics. Supervisor: Dr. Gustavo Turecki.
- Generating and analyzing a VisiumHD dataset for multi-omic spatial characterization of Major Depressive Disorder in the human Prefrontal Cortex.
- Analyzing RNAseq and EMseq data of pyramidal neurons from prefrontal cortex of depressed suicides with history of childhood abuse.
- Studying paraventricular thalamus using snRNAseq in the context of childhood abuse.

Genome Institute of Singapore (GIS), Research Officer,
Singapore

Jul 2022 – Jul 2025

- Supervisor: Dr. Jinyue Liu.
- Preparing a first-author paper on temporal and spatial characterization of a human midbrain-like organoid model of cellular vulnerability in Parkinson's disease.
- Analyzed in-house and publicly available scRNAseq and MERFISH spatial transcriptomics data.
- Studied glioblastoma multiforme tumor biopsies and ligand–receptor interactions between invasive tumor subtypes and their immediate cellular neighbourhoods using CosMx spatial transcriptomics.
- Characterized cell-type-specific differential expression effects caused by *Smchd1* mutation in embryonic gonads of mice in collaboration with Dr. Xue Shifeng's lab at NUS.

Yale-NUS College, Undergraduate Researcher, Singapore

Aug 2021 – May 2022

- Supervisor: Dr. Jan Gruber.
- Wrote bachelor's thesis titled *Cosmetic neurology at scale: developing a high-throughput phenotypic screening for novel nootropics in D. melanogaster*.
- Created custom Arduino behavioural set-up for vibrational conditioning, CAFÉ feeding assay using capillaries, and a hypoxia recovery assay.

Brigham and Women's Hospital, Summer Research Intern,
Online

May 2021 – Sept 2021

- Supervisor: Dr. William Renthall.
- Studied cell-type-specific enhancers in dorsal root ganglion.
- Performed analysis of scRNAseq and complementary scATACseq datasets in mice.

Institute of Molecular and Cell Biology (IMCB), Intern,
Singapore

May 2020 – May 2021

- Supervisor: Dr. Ajay Mathuru.
- Ran behavioural tests investigating the role of oxytocin receptors in scotophobic and thigmotaxic behaviour using OpenCV-based analysis.

Yale-NUS College, Lab Assistant, Singapore

Sept 2019 – Dec 2019

- Supervisor: Dr. Philip Johns.
- Extracted over 100 RNA samples for a study investigating aggressive behaviour in *t. discus* flies.

Duke-NUS Centre for Computational Biology, Intern,
Singapore

May 2019 – Jul 2019

- Supervisor: Dr. Steven Rozen.

- Evaluated the performance of software used in mutational signature analysis (e.g. SigProfiler, SignatureAnalyzer, signeR, deconstructSigs).
- Formatted variant calling files (VCFs) for downstream analysis.

Volunteering Experience

Academic Buddy LT, Student Mentor (online) Aug 2018 – Present

Singapore Society for Neuroscience (SfN.SG), Newsletter Editor and Web Master, Singapore Mar 2023 – Jul 2025

Integrative Neuroscience Association, Staff Writer, Lithuania Dec 2020 – Mar 2023

Computer Skills and Languages

Languages: Lithuanian (Native), English (Fluent), French (Intermediate, B1), Chinese (Elementary).

Technical skills: RStudio (Advanced), Python (Intermediate), OCaml (Elementary), C++ (Elementary).

Interests: DJing, classical guitar, basketball, snowboarding.